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2026-03-21

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1: BZ Belousov, Zhabotinsky, Prigogine 2
2: Turing 1952 2
3: Leibler, Fredrickson-Helfand 2
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. 4
. 4

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- : 5
- : 5 4 1

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1 Gibbs, Volmer-Weber, Becker-Döring

2 Michaelis-Menten, Langmuir-Hinshelwood

3		Leibler, Fredrickson-Helfand
4	BZ	Belousov, Zhabotinsky, Prigogine
5		Turing, Meinhardt-Gierer
6		Lehn, Whitesides
7		Semenov, Hinshelwood
8	-	Flory, Stockmayer
9	RNA world	Miller-Urey, Gilbert
10		Goldbeter, Prigogine

1: BZ Belousov, Zhabotinsky, Prigogine

- BZ Belousov-Zhabotinsky Belousov 1951 1959 Zhabotinsky 1960 1968 Field, Körös, Noyes 1972 Field-Noyes 1974 3 Oregonator Winfree 1972
- Prigogine 1977 Prigogine
- : Oregonator Hopf Brusselator Prigogine-Lefever 1968
- : → → → + →Hopf
- : Hopf Prigogine

2: Turing 1952

- Turing 1952 activator inhibitor Turing Meinhardt-Gierer 1972, 2000 40 Castets et al. 1990 CIMA Kondo-Miura 2010
- : activator inhibitor inhibitor activator
- : activator-inhibitor
- : activator-inhibitor —— ——

3: Leibler, Fredrickson-Helfand

- AB A B Leibler 1980 Fredrickson-Helfand 1987
- : f N × q *S(q) q* q* N f
- : q* A/B
- : q* A/B N f D

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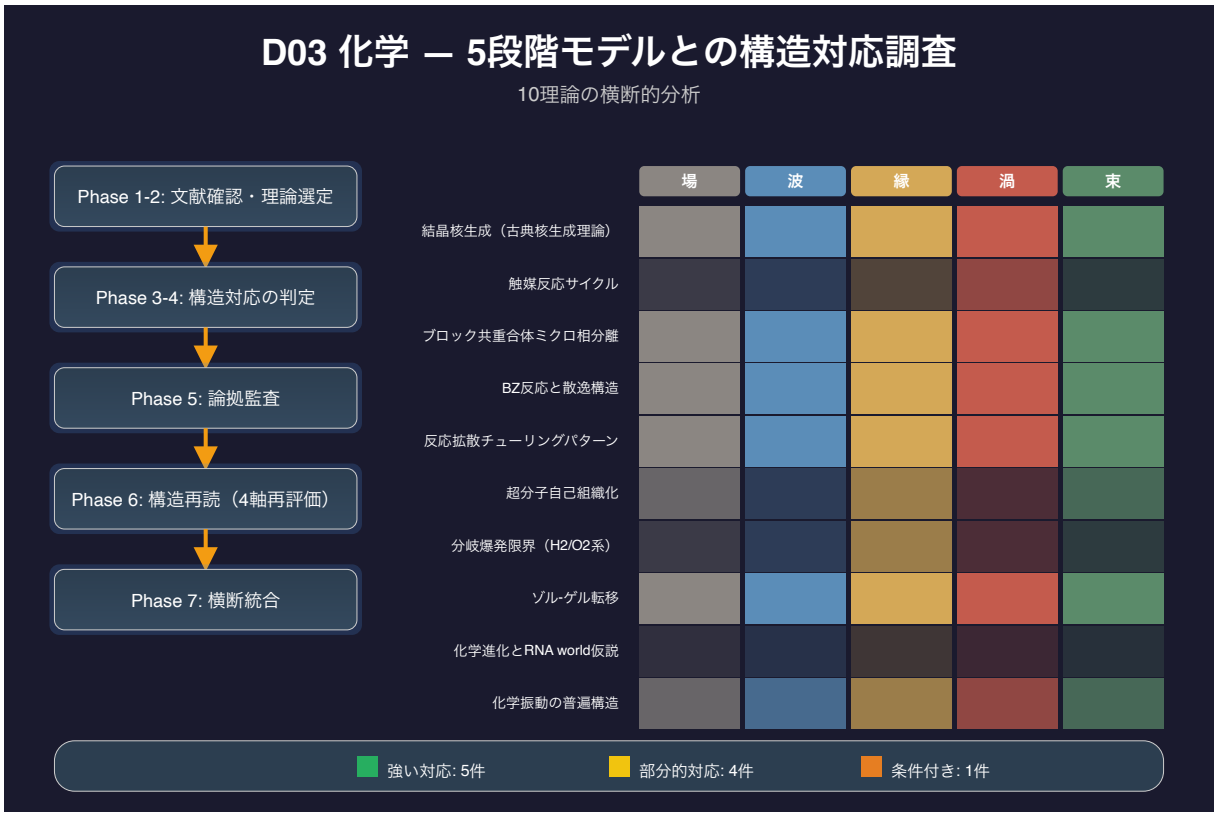
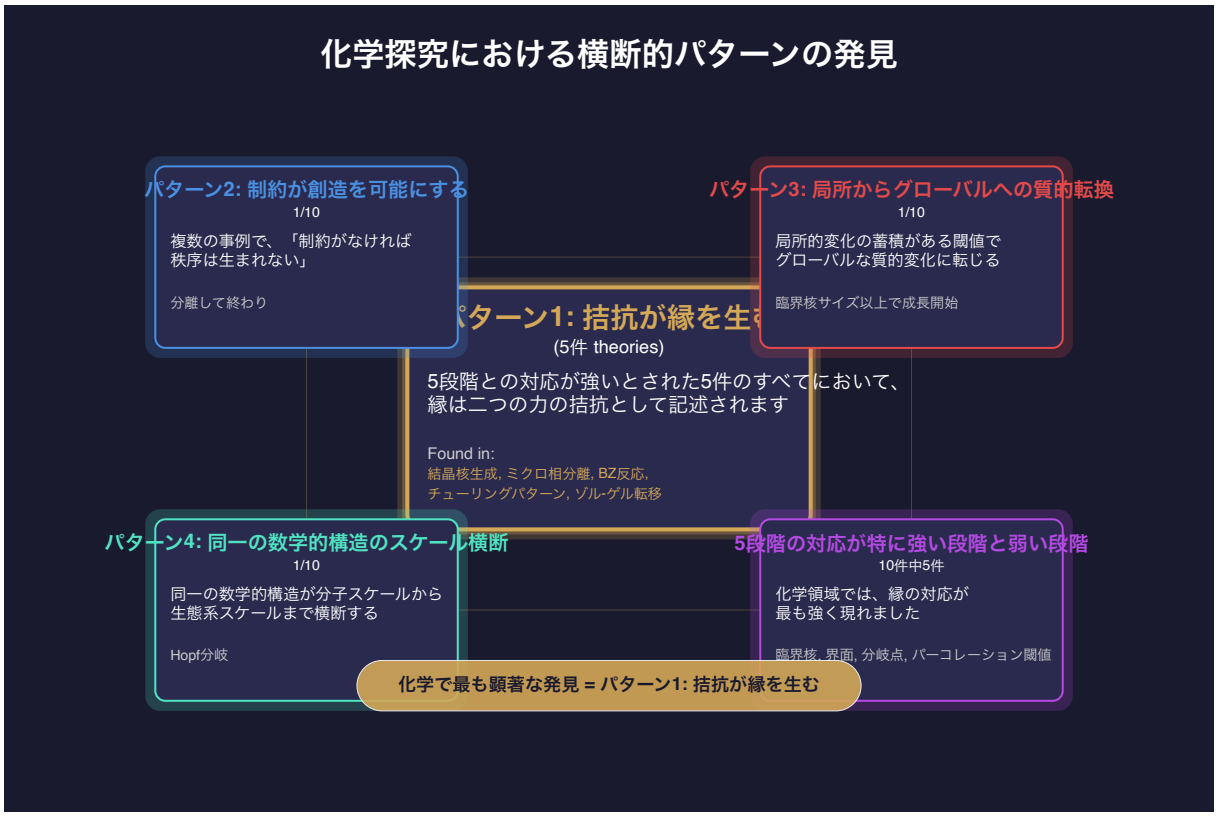


1: Domain —

2: Domain — ×5



3: Domain —

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- 5.1 → 5
- Ca^2 Berridge 1997
- 5 —— ——
- RNA world

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